

INTRODUCTION

As the part of the B.Tech curriculum under MG University, we, the Fourth Semester Electrical and Electronics Engineering students of Adi Shankara Institute of Engineering & Technology, Kalady have visited Diesel Power Plant, Nallalam on 10th January 2014 . The IV was successfully completed under the able guidance of Jasmy miss and Alan sir.

This training gave us a rare opportunity to understand the functioning of Asia's 2nd largest diesel power plant and experience power production management at very large scale. The controlling of power plant is fully computerized and semiautomatic.

The Kozhikode Diesel Power Plant, established in 1999 is Asia's second largest Diesel Power Plant with a plant capacity of 128 MW. It is a semi-automatic plant using a Diesel Engine as its prime mover.

FEATURES

- Plant Capacity of 128 MW
- 8WARTSILA engines each of 16 MW
- Low Sulphur Heavy Stock (LSHS) is used as permanent fuel
- 5 generating transformers each of 38MVA used
- Total power generated : 18.9 MW
- Frictional power : 2.439MW
- Net electrical power : 16MW
- Fuel consumption : 0.87 Kg/sec
- Thermal efficiency : 52.5 %
- Brake thermal efficiency : 45.81 %

MAIN UNITS OF POWER PLANT

DIESEL ENGINE

The heart of the plant is the diesel engine. The diesel engine is made by the Finland Company WARTSILA. The engine employed here is 4-stroke, turbocharged, intercooled engine with direct twin fuel injection.

Engine Specification

- 18V 46C WARTSILA engines
- Stroke length : 580mm
- Swept volume : 96.4 l per cylinder
- No. of valves : 4 (2 inlets, 2 exhausts)
- Cylinder configuration : 18 cylinders in V form
- V-angle : 45°
- Compression ratio : 14:1
- Direction of the rotation : clockwise
- Speed : 500 rpm
- Firing pressure in the cylinder : 500 bar
- Charge air pressure : 3.1 bar
- Bore diameter : 460mm
- Piston diameter : 460mm

Cylinder Configuration

The engine consists of 18 cylinders of bore diameter of 460mm each and also the cylinders are arranged in 'V' shape. Hence it is called 18V 46C. The cylinders are arranged in two banks:- A bank and B bank.

- A-bank, which is also called the maneuvering side, bears the operating devices like start/stop instrument panel.

- B-bank, which is also called the rear side of the engine, is the driving end of flywheel end of the engine. The turbocharger is located at this end.

The complexities of the diesel engine used here points to certain necessary initial conditions which have to be strictly satisfied before its operation. 18 necessary initial conditions that have to be satisfied for the operation of the diesel engine are:

1. Lube oil inlet pressure > 5 bar
2. Fuel oil inlet pressure > 2 bar
3. Start air pressure > 18 bar
4. HT water A-bank outlet temperature > 60°C
5. HT water B-bank outlet temperature > 60°C
6. Turning gear disengaged
7. Mechanical over speed shutdown inactive.
8. Stop valve closed or auto stop inactive.
9. Generator earthing switch open.
10. Outgoing feeder earthing switch open.
11. Generator bearing oil flow LO pumps run.
12. Generator Cubicle MCB open.
13. Protection relay fault inactive.
14. Breaker trip alarm inactive.
15. Shut down alarm inactive.
16. Engine stopped.
17. Power plant emergency stop inactive.
18. CFC081 emergency stop inactive.

Alternator and exciter

The diesel engine is directly coupled to vertical shaft alternator having a capacity of 16MW. The alternator is 12 pole salient pole synchronous generators which has star connecting stator winding with double layer lap winding. The stator winding has 14 slots/pole forming it 168 slots in total.

Alternator specification

- KVA rating : 18824 KVA
- KW rating : 16 MW
- Rating voltage : 11 KV
- Rated current : 988 A
- Power factor : 0.85
- Rated frequency : 50 Hz
- Rated speed : 500 rpm
- Excitation current : 405 A
- Excitation voltage : 146 V
- Ambient temperature : 45°C

The exciter is used for supplying the excitation voltage to the main field by induction method. The exciter used here is a separately excited brushless exciter. The stator of the exciter is provided with a 76V DC. An alternating emf is generated when the rotor rotates in the stator field. The rotor is star connected and is coupled to the alternator shaft. A rectified output of this alternating EMF is transmitted to the main field of the alternator.

Exciter specification

- KW rating : 59.1 KW
- Excitation ampere : 9.1 A
- Excitation volt : 76 V
- Connection : star connection
- Volt : 146 V DC
- Ampere : 405 A
- Speed : 500 rpm
- Frequency : 66.7 Hz

GENERATOR TRANSFORMER

- 5 GT units each of 38MVA, rated 11/110KV.

- Each GT provided with output of two engines.
- One of the five is for backup purpose.

STATION TRANSFORMERS

- Step down transformers of rating 11KV/433V.
- Supplies engine axillaries.
- Four 2MVA and two 250 KVA transformers.
- The 2MVA transformers are connected to MV bus bar.
- The 250KVA transformers operate during plant shut down period.

MV ROOM

Medium voltage switch gear monitors the 11KV supply of the plant. The Generator Incomer Breaker of the MV room is connected with 16 MW Generators via three underground cables. 4 MV buses are installed in MV Room having 5 MV switch gears in each

- Bus PT
- Generator Incomer 1
- Generator Incomer 2
- Generator transformer
- Station transformer

Bus voltage is measured using Bus PT. Generator Incomer takes the supply from the two generators to the MV bus. Generator transformer breaker takes the supply from MV bus to GT. Station transformer breaker takes the supply from MV bus to ST.

VCB (Vacuum Circuit Breaker) is used as MV switch gears. There are two types of springs in breakers

- Closing Spring: It is charged using a DC motor.

- Tripping spring: It is charged while closing spring discharges. VCBs are 12KV 50Hz breakers with a pf withstand of 28 KV. These breakers can be operated in both remote and manual position.

LIGHTING ARRESTOR AND SURGE COLLECTOR PANEL

The LA and SC units are connected to Generator Incomer. When lightning stroke is incident on the grid, LV and HV side of GT is slightly energized. HV side discharges through LA connected to grid near GT and LV side discharges through the LA inside the LA and SC panel to protect generator winding.

SC panel can be considered as a capacitor having high impedance value. At normal conditions there is no electric flow through SC unit as impedance is high. When surge current (high frequency current) develops the capacitive impedance reduces considerably. Hence surge current is discharged SC unit to ground.

LV ROOM

The 11 KV supply from MV is step down to 433 V by means of a station transformer and it is provided to LV room. All the auxiliaries are supplied from the LV room. Each and every auxiliary module in such a way that for each engine the control unit are accommodated on separate Motor Control Cubicle (M C C). MCC can be explained as follows,

- There is a total of 10 MCC installed.
- MCCs 1-8 are specified for each engine.
- MCC 9, being an important MCC feeds all the booster unit, battery charges feeder, boiler feeder, starting air unit, and emergency lighting.

MCC 9 is always provided with a supply, as supply failure of MCC 9 result in plant shut down.

BLACK OUT AND BLACK START GENERATOR

When all the supply from the station to the plant fails (grid failure), the condition is said to be “Black Out”. In such conditions a small diesel generator called Black Start Generator of 500 KVA is used for maintaining power supply to the plants. Black start generator energizes MCC 9 there by promoting plant production.

LV room consists of 55 cells of 2 volts which provides 110 V DC power back. Air circuit breakers are used in LV room which are basically classified into two based on their current carrying capacity.

- 3200 Ampere: ACB of frequency 50-60 Hz, ambient temperature of 45°C
- 1250 Ampere: ACB of frequency

SUMMARY

KDPP is one of the major power plants in North Kerala. It forms the main power backup solution for the increasing demands. Being a diesel plant, it has to assure perfect cleanliness. The plant is designed for continuous working as per international standards. The plant setup is done by WARTSILA NSD, Finland. All the important parts, the working and maintenance of the plant have been understood.